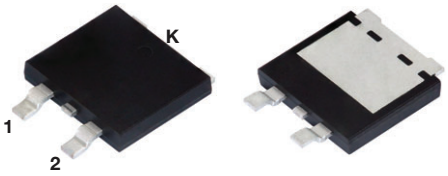
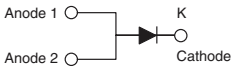
Dual High-Voltage TMBS® (Trench MOS Barrier Schottky) Rectifier

Ultra Low V_F = 0.52 V at I_F = 5 A

eSMP® Series
SMPD (TO-263AC)



Top View Bottom View



FEATURES

- Trench MOS Schottky technology generation 2
- Very low profile - typical height of 1.7 mm
- Ideal for automated placement
- Low forward voltage drop, low power losses
- High efficiency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
  - Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE Available

RoHS COMPLIANT HALOGEN FREE

LINKS TO ADDITIONAL RESOURCES

3D Models

TYPICAL APPLICATIONS

For use in high frequency DC/DC converters, switching power supplies, freewheeling diodes, OR-ing diode, and reverse battery protection in commercial, industrial, and automotive application.

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	20 A
V_{RRM}	120 V
I_{FSM}	150 A
V_F at I_F = 20 A (T_A = 125 °C)	0.71 V
T_J max.	175 °C
Package	SMPD (TO-263AC)
Circuit configuration	Single

MECHANICAL DATA

Case: SMPD (TO-263AC)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant
Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 and HM3 suffix meets JESD 201 class 2 whisker test

Polarity: as marked

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**V20DM120**	**UNIT**
Maximum repetitive peak reverse voltage	V_{RRM}	120	V
Maximum DC forward rectified current (fig. 1)	$I_{F(AV)}$ (1)	20	A
	$I_{F(AV)}$ (2)	5.5	
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	150	A
Operating junction and storage temperature range	T_J, T_{STG}	-40 to +175	°C

Notes
(1) With infinite heatsink
(2) With recommended pad size, 2 oz FR4 PCB

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage per diode	I_F = 5 A	T_A = 25 °C	V_F (1)	0.62	-	V
	I_F = 10 A			0.77	-	
	I_F = 20 A			1.02	1.1	
	I_F = 5 A	T_A = 125 °C		0.52	-	
	I_F = 10 A			0.61	-	
	I_F = 20 A			0.71	0.79	
Reverse current at rated V_R per diode	V_R = 90 V	T_A = 25 °C	I_R (2)	0.01	-	mA
		T_A = 125 °C		3	-	
	V_R = 120 V	T_A = 25 °C		-	0.8	
		T_A = 125 °C		5	15	

Notes
(1) Pulse test: 300 µs pulse width, 1 % duty cycle
(2) Pulse test: Pulse width ≤ 5 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**V20DM120**	**UNIT**
Typical thermal resistance	$R_{\theta JC}$	1.6	°C/W
	$R_{\theta JA}$ (1)(2)	48	

Notes
(1) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$ - junction-to-mount
(2) Free air, without heatsink

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
V20DM120-M3/I	0.55	I	2000/reel	13" diameter plastic tape and reel
V20DM120HM3/I (1)	0.55	I	2000/reel	13" diameter plastic tape and reel

Note
(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

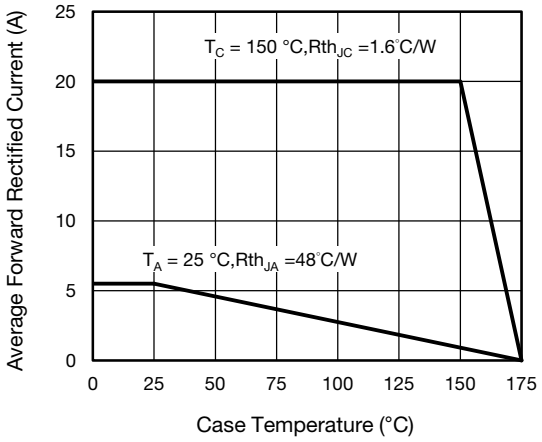


Fig. 1 - Forward Current Derating Curve

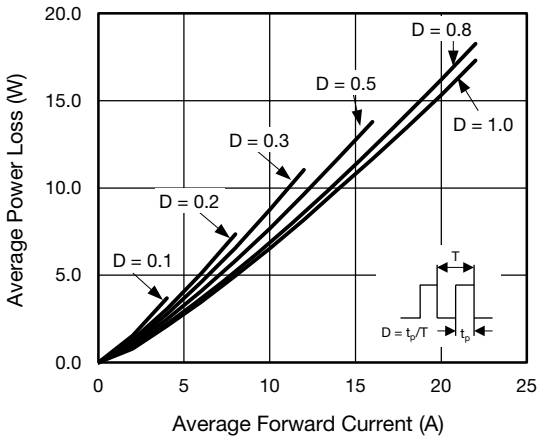


Fig. 2 - Average Power Loss Characteristics

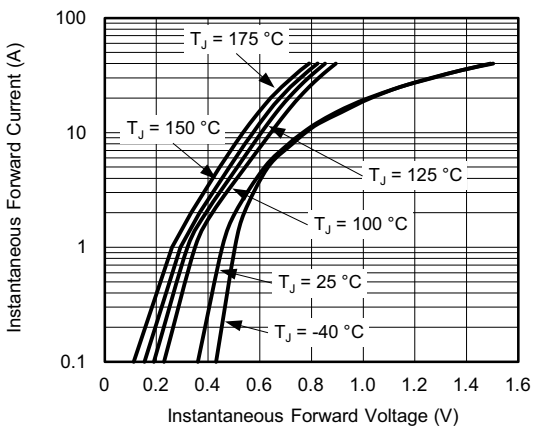


Fig. 3 - Typical Instantaneous Forward Characteristics

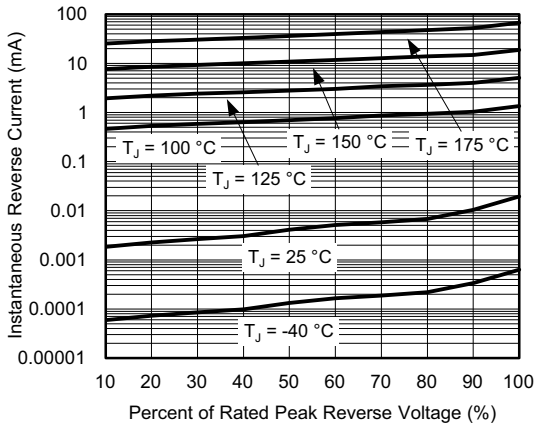


Fig. 4 - Typical Reverse Characteristics

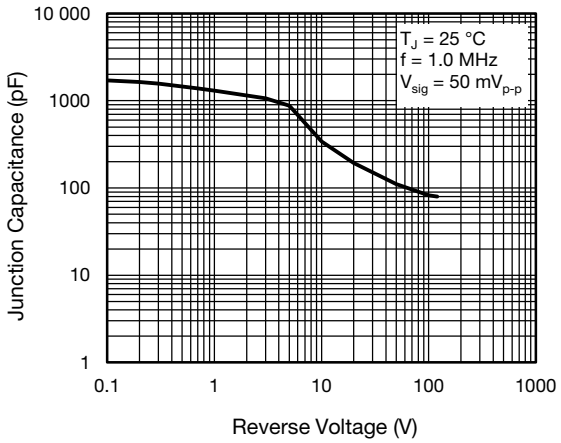


Fig. 5 - Typical Junction Capacitance

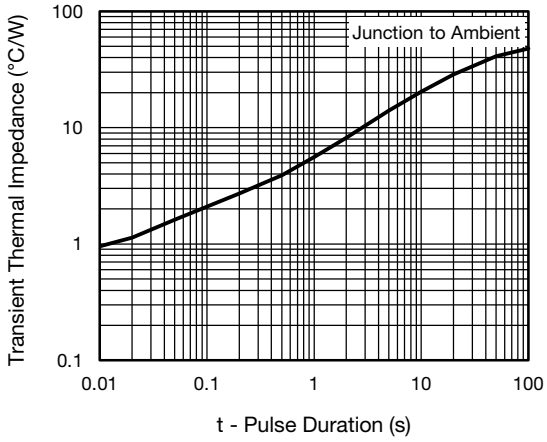


Fig. 6 - Typical Transient Thermal Impedance

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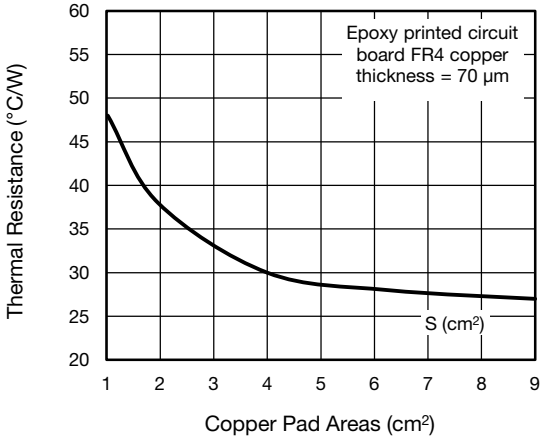
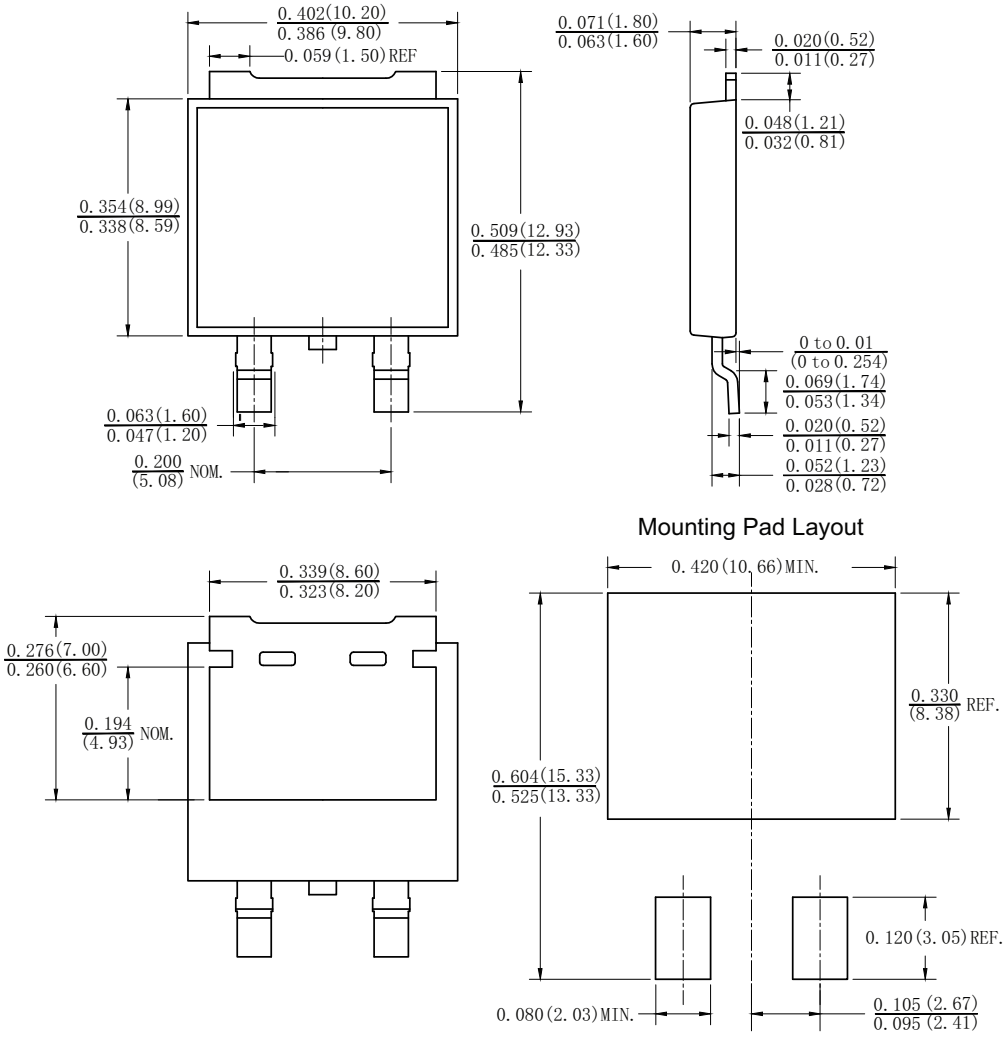


Fig. 7 - Thermal Resistance Junction-to-Ambient vs. Copper Pad Areas

PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMPD (TO-263AC)



Mounting Pad Layout

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